

Keyword Research and Competitive Analysis Using SEO Tools

EX NO: 1

Date:

Aim:

To perform keyword research and competitor analysis using SEO tools.

Objectives

- Identify relevant keywords.
- Analyze competitor websites.
- Compare keyword metrics.
- Prepare a simple content plan.

Tools Required

- Google Keyword Planner
- Ubersuggest
- SEMrush/Ahrefs
- Google Trends
- MS Excel or Google Sheets

Theory

Keyword research identifies words users search for online. Competitive analysis studies competitor websites to find their keywords, rankings, and SEO strategies. This helps select effective keywords and create better content for higher search engine rankings.

Example

Keyword	Search Volume	Difficulty
SEO Tips	15,000	Medium
Keyword Research	10,000	Low
SEO Tools	20,000	Medium

Example

Website: Online Book Store

Keyword	Search Volume	Difficulty
SEO Tips	15,000	Medium
Keyword Research	10,000	Low
SEO Tools	20,000	Medium

Procedure

1. Open an SEO tool.
2. Enter a seed keyword.
3. Collect keyword suggestions.
4. Check search volume and difficulty.
5. Analyze top competitors.
6. Select suitable keywords.
7. Prepare a content plan.

Sample Content Plan

Content Title	Primary Keyword
SEO Basics	SEO Tips
Keyword Research Guide	Keyword Research
Best SEO Tools	SEO Tools

Observation

- Long-tail keywords have lower competition.
- High-quality competitor content ranks better.
- Search intent affects keyword selection.

Result

Keyword research and competitor analysis were completed successfully. Suitable keywords were selected, and a basic content plan was prepared to improve SEO performance.

On-Page Optimization for Blog Posts and Web Pages

EX NO: 2

Date:

Aim:

To optimize a blog post or web page using on-page SEO techniques.

Objectives

- Learn on-page SEO basics.
- Optimize titles, headings, keywords, and images.
- Improve search engine ranking and readability.

Tools Required

- Google Chrome
- Google Keyword Planner
- MS Word/Google Docs
- Yoast SEO or Rank Math
- Google Search Console

Theory

On-page SEO is the process of optimizing webpage content and HTML elements to improve search engine rankings. It includes optimizing the title, meta description, URL, headings, keywords, images, and internal links to make content more visible and user-friendly.

Example

Topic: Digital Marketing Benefits

- **Keyword:** Digital Marketing Benefits
- **Title:** 10 Benefits of Digital Marketing for Small Businesses
- **URL:** /digital-marketing-benefits
- **Meta Description:** Learn how digital marketing helps small businesses grow online.

Procedure

1. Select a topic.
2. Research keywords.
3. Create an SEO-friendly title and URL.
4. Write content using headings and keywords.
5. Add optimized images and links.
6. Check SEO score and publish.

Sample Content Plan

Item	Example
Topic	Digital Marketing Benefits
Primary Keyword	Digital Marketing Benefits
Title	10 Benefits of Digital Marketing
URL	/digital-marketing-benefits
MetaDescription	Benefits of digital marketing for businesses.

Observation

- Keywords were used naturally.
- Title, headings, and URL were optimized.
- Images included ALT text.
- Readability and SEO score improved.

Result

The webpage was successfully optimized using on-page SEO techniques, improving search engine visibility, content quality, and user experience.

Backlink Analysis and Link-Building Strategies

EX NO: 3

Date:

Aim:

To analyze website backlinks and implement effective link-building strategies to improve SEO performance.

Objective

- Analyze the backlink profile of a website.
- Identify quality and toxic backlinks.
- Compare competitor backlinks.
- Develop ethical link-building strategies.

Tools Required

- Google Search Console
- Ahrefs
- SEMrush
- Moz Link Explorer
- Ubersuggest
- Microsoft Excel / Google Sheets

Theory

Backlinks are links from one website to another and are an important SEO ranking factor. High-quality backlinks improve domain authority, search rankings, and website traffic. Common link-building methods include guest blogging, content marketing, broken link building, and directory submissions.

Example: Website: <http://www.greenleaforganics.com>

Metric	Value
Backlinks	1,250
Referring Domains	180
Domain Authority	42
Toxic Links	20

Procedure

1. Open a backlink analysis tool (e.g., Ahrefs or Google Search Console).
2. Enter the website URL.
3. Analyze backlinks, referring domains, and toxic links.
4. Compare competitor backlink profiles.
5. Plan and implement suitable link-building strategies.
6. Monitor backlink growth regularly.

Sample Content Plan

Week	Content	Activity
1	Blog Post	Guest Posting
2	Infographic	Social Sharing
3	Case Study	Resource Page Links
4	How-to Guide	Broken Link Building

Observation

- High-quality backlinks improved website authority.
- Toxic backlinks reduced SEO performance.
- Competitor analysis revealed new backlink opportunities.

Result

Backlink analysis helped evaluate the website's SEO strength and identify improvement areas. Ethical link-building strategies increased domain authority, search visibility, and organic traffic.

Technical SEO Audit and Implementation

EX NO: 4

Date:

Aim:

To perform a technical SEO audit and improve website performance.

Objective

- Identify technical SEO issues.
- Improve crawling, indexing, and page speed.
- Enhance website performance.

Tools Required

- Google Search Console
- Google PageSpeed Insights
- Screaming Frog SEO Spider
- Google Lighthouse
- GTmetrix

Theory

Technical SEO improves a website's crawlability, indexing, speed, mobile usability, HTTPS, and overall search engine performance.

Example

Issue: Slow page speed and broken links.

Solution: Optimize images, fix broken links, and submit an XML sitemap.

Procedure

1. Analyze the website.
2. Check page speed and mobile usability.
3. Find SEO issues using audit tools.
4. Fix errors (links, metadata, sitemap).
5. Test and monitor improvements.

Sample Content Plan

Audit Item	Action
Page Speed	Optimize images
Broken Links	Fix URLs
XML Sitemap	Create & submit
Meta Tags	Add missing tags

Observation

Technical issues were identified and corrected, improving website performance.

Result

The website achieved better crawlability, faster loading, and improved SEO performance.

Monitoring and Improving Page Speed for SEO Performance

EX NO: 5

Date:

Aim:

To monitor and improve website page speed for better SEO performance.

Objective

- Measure page speed.
- Identify performance issues.
- Apply speed optimization techniques.
- Improve SEO and user experience.

Tools Required

- Google PageSpeed Insights
- GTmetrix
- Google Lighthouse
- Chrome Browser

Theory

Page speed is the time a webpage takes to load. Faster websites improve user experience, reduce bounce rate, and achieve better search engine rankings.

Example

A webpage load time improved from **5.5 s** to **2.2 s** after image compression and code optimization.

Procedure

1. Test the website using PageSpeed Insights.
2. Identify speed issues.
3. Optimize images and minify CSS/JavaScript.
4. Enable browser caching.
5. Retest and compare the results.

Sample Content Plan

Item	Details
URL	www.example.com
Initial Speed	5.5 s
Optimization	Image compression, Minification, Caching
Final Speed	2.2 s

Observation

- Faster page loading.
- Higher performance score.
- Better user experience.

Result

The webpage speed was improved successfully, resulting in enhanced SEO performance and faster website loading.